

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Own technical direction for GPU-accelerated AI/ML platform infrastructure: provisioned multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA and Terraform-managed IaC, balancing scalability and cost for training and inference workloads.
- Standardized platform patterns for vector databases (Pinecone, FAISS), model serving, and RAG/LLMOps orchestration on Kubernetes, with GitHub Actions CI/CD and a deployment-gated pipeline enforcing quality, latency, and drift checks.
- Built observability and autoscaling for GPU inference serving using Prometheus and Grafana; lead design reviews and mentor engineers on infrastructure standards across the AI platform team.

QureAI

Mar 2026 – Present

ML Infrastructure Engineer

- Designed and implemented a Neo4j knowledge graph and vector-search retrieval layer deployed to production, standardizing infrastructure patterns for model registries and orchestration frameworks in a regulated environment.
- Cut p99 LLM inference latency 420ms to 140ms across 50K+ daily requests by profiling and migrating synchronous PyTorch-backed serving to async vLLM, optimizing GPU throughput under autoscaling.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Built and operated 25+ production C++ and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production performance and eval tooling with rubric-based scoring, drift detection, and per-call latency benchmarking across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Java, C++, Go, SQL

AI/ML Platform: MLOps/LLMOps, RAG Pipelines, Vector Databases (Pinecone, FAISS, Neo4j), Model Serving, PyTorch, vLLM, SageMaker

GPU & Infra: NVIDIA/CUDA, Multi-Cloud GPU Orchestration (AWS, GCP, CoreWeave), Terraform, Kubernetes, Docker, CI/CD (GitHub Actions)

Observability & Ops: Prometheus, Grafana, Autoscaling, Fault Tolerance, Distributed Systems, Technical Mentorship

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer